Lab 3: Probability and Distributions

Instructor notes – Part 1: prep reading, Part 2: podium/circulating cheat sheet

This single file has two parts, both instructor-only (not a student handout). **Part 1** is read beforehand – an answer key and the reasoning behind each Act. **Part 2** (after the page break, large print) is the copy to have open while circulating during lab.

Schedule note: Labor Day cancels this week's Monday meeting, so unlike a normal week, this lab now runs with **no prior lecture at all** – Wednesday's probability/Bayes' rule lecture comes *after* this lab, not before, and the bestiary of distributions has moved to next Monday (Week 4). That puts this lab in a similar position to `NOTES_Lab1_MathRefresher.Rmd` (content ahead of any formal instruction), except here it's a new concept (Bayes' rule) rather than rusty-but-familiar calculus, so groups may need more up-front framing than usual – consider a short (5 min) live walkthrough of conditional probability notation before releasing Act 1, since the Act itself doesn't re-derive Bayes' rule from scratch. This doc exists mainly as a quick-reference answer key for spot-checking groups, kept brief.

Act 1: camera-trap classifier PPV

Same false-positive/PPV technique as the medical-testing example Wednesday's lecture will formalize (Bolker 4.3.1), different numbers ($P(\text{species}) = 0.0001$, sensitivity $= 0.98$, false positive rate $= 0.01$) – since the lecture hasn't happened yet, Act 1 is students' first encounter with this idea, not a repeat.

```r
p_S <- 1e-4; sens <- 0.98; fpr <- 0.01
p_notS <- 1 - p_S
p_F_and_S    <- sens * p_S
p_F_and_notS <- fpr * p_notS
p_F  <- p_F_and_S + p_F_and_notS
PPV  <- p_F_and_S / p_F
c(P_F_and_S = p_F_and_S, P_F_and_notS = p_F_and_notS, P_F = p_F, PPV = PPV)
```

```
##    P_F_and_S P_F_and_notS          P_F          PPV
##  0.000098000  0.009999000  0.010097000  0.009705853
```

Answer key: $P(F \cap S) = 0.000098$, $P(F \cap \neg S) = 0.009999$, $P(F) \approx 0.0101$, PPV ≈ 0.0097 ($\approx 1\%$). The "negligible term" approximation ($P(F) \approx 0.01$, dropping the S-contribution) gives essentially the same answer – a group that uses the shortcut isn't wrong.

Natural-frequency table (N = 100,000), rounded:

	Flagged	Not flagged	Total
Actually the species	≈ 10	≈ 0	10
Not the species	$\approx 1{,}000$	$\approx 98{,}990$	99,990
Total	$\approx 1{,}010$	$\approx 98{,}990$	100,000

PPV $= 10/1010 \approx 0.0099$ – matches Part 1.

Watch for: groups who compute $P(S \mid F)$ by dividing by $P(S)$ instead of $P(F)$ (the classic $P(A \mid B)$ vs. $P(B \mid A)$ mix-up – this is exactly the base rate fallacy Wednesday's lecture will name explicitly). If a group's PPV comes out > 0.5, that's almost always the tell.

Act 2: binomial detections, and when Poisson is a good stand-in

$N = 8$, $p = 0.3$, $k = 3$.

```
N <- 8; p <- 0.3; k <- 3
choose(N, k) * p^k * (1 - p)^(N - k)
```

```
## [1] 0.2541218
```

```
dbinom(k, N, p)
```

```
## [1] 0.2541218
```

Answer key: $P(X = 3) = 0.2541$. Full distribution mode is at $k = 2$ (check with `which.max(dbinom(0:8,8,0.3))`).

Poisson comparison, part A ($N = 8, p = 0.3$ vs. $\lambda = 2.4$): binomial variance $Np(1-p) = 1.68$; Poisson variance $\lambda = 2.4$ – mismatched, and the two barplots should look visibly different (Poisson has a fatter right tail).

Poisson comparison, part B ($N = 100, p = 0.024$ vs. $\lambda = 2.4$): binomial variance $= 100(0.024)(0.976) = 2.3424$ – close to 2.4, and the two barplots should look nearly identical. Rule of thumb (2f): *Poisson is a good stand-in for binomial when N is large and p is small* (so $(1-p) \approx 1$ and $Np(1-p) \approx Np = \lambda$).

Watch for: groups who conclude Poisson is "just always fine" from part B alone without noticing part A disagreed – push them back to the variance mismatch in part A as the reason it isn't universal.

Act 3: continuous body mass ($\mu = 4.5$, $\sigma = 0.6$)

```r
mu <- 4.5; sigma <- 0.6
1 - pnorm(5.5, mu, sigma)
```

```
## [1] 0.04779035
```

```r
qnorm(0.9, mu, sigma)
```

```
## [1] 5.268931
```

Answer key: $P(\text{mass} > 5.5) \approx 0.048$ ($\approx 5\%$); the heaviest-10% cutoff is ≈ 5.27 kg.

Watch for: 1b/1c (density vs. probability) is the conceptual crux of this Act – a group that answers "$P(\text{mass} = 4.5)$ is very small but not zero" hasn't fully separated density from probability yet. Point them back at the shaded-area picture from 1a: a single point has zero *width*, hence zero area, hence zero probability, regardless of how tall the curve is there.

PART 2 – CHEAT SHEET (circulating)

Copy for circulating during lab. Full reasoning and code: see Part 1 above.

Act 1 – classifier PPV

- $P(F \cap S) = 0.000098$, $P(F \cap \neg S) = 0.009999$, $P(F) \approx 0.0101$
- PPV ≈ 0.0097 ($\approx 1\%$)
- Freq. table check (N=100k): $10/1010 \approx 1\%$ – matches
- **Watch for:** PPV > 0.5 usually means $P(A|B)/P(B|A)$ flipped

Act 2 – binomial vs. Poisson

- $N = 8, p = 0.3, k = 3$: hand calc and `dbinom(3,8,0.3)` both $= 0.2541$
- Mode of full distribution: $k = 2$
- Var mismatch ($N = 8, p = 0.3$): binom 1.68 vs. Pois 2.4 – **don't match**
- Var match ($N = 100, p = 0.024$): binom 2.3424 vs. Pois 2.4 – **do match**
- Rule of thumb: Poisson $\approx$ binomial when N large, p small

Act 3 – continuous mass

- $P(\text{mass} > 5.5) \approx 0.048$; top-10% cutoff ≈ 5.27 kg
- **Watch for:** "small but nonzero" answers to $P(X = 4.5)$ – push back to the shaded-area picture (zero width $\rightarrow$ zero probability)